03 — Forecast and Position Analytics

ARCM: Adaptive Rate Curve Model | Liyuan Zhang

Explores the LSTM multi-horizon forecast and position-level P&L, risk metrics, and scenario analysis for CUSIP 912797UE5.

Prerequisite: Full pipeline stack:

```
python pipeline.py
python regime.py
python dynamics.py
python forecast.py --epochs 200
python position.py
```

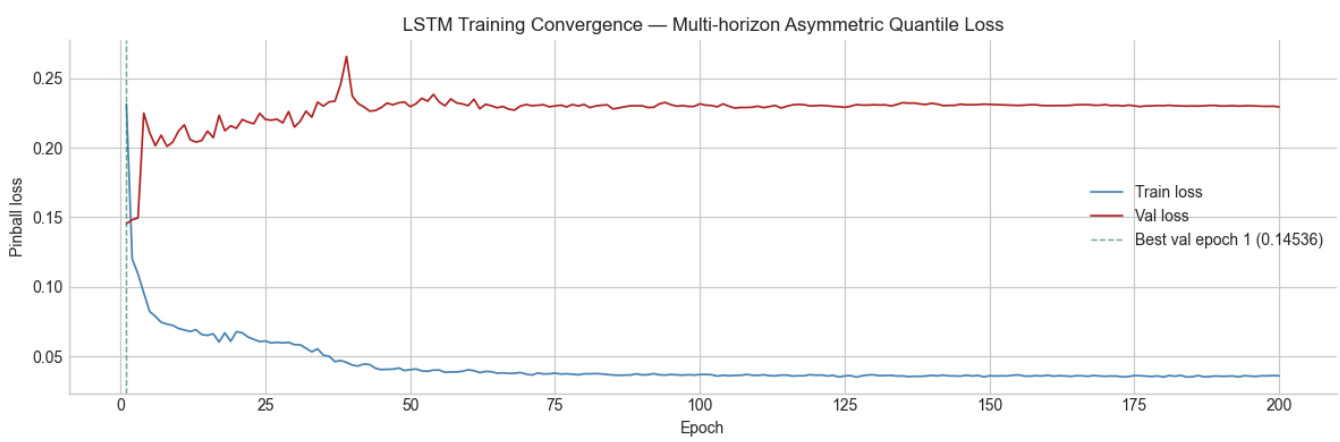
1. Load outputs

```
Forecast distribution : 503 rows
Position report       : 6 rows
```

	q10	q50	q74
0	2.49179	3.053973	3.340099

2. Training convergence

Multi-horizon training (21/63/126/252d simultaneously) gives ~40x more training signal vs single-horizon. Best val loss is epoch 1 — a known limitation of the 2800-row post-ZLB dataset. Improves as data accumulates.

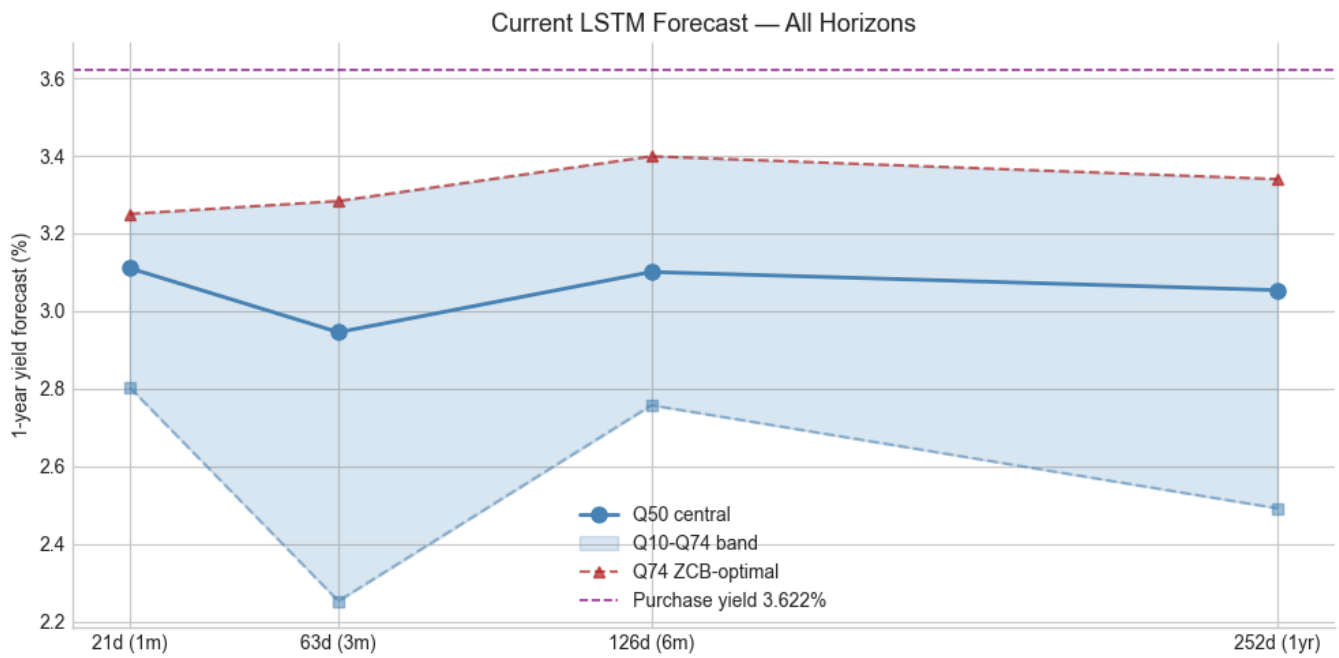



```
Best val loss : 0.14536 at epoch 1
Final train   : 0.03593
Train/val gap : 0.2x
```

3. Out-of-sample forecast distribution vs actuals

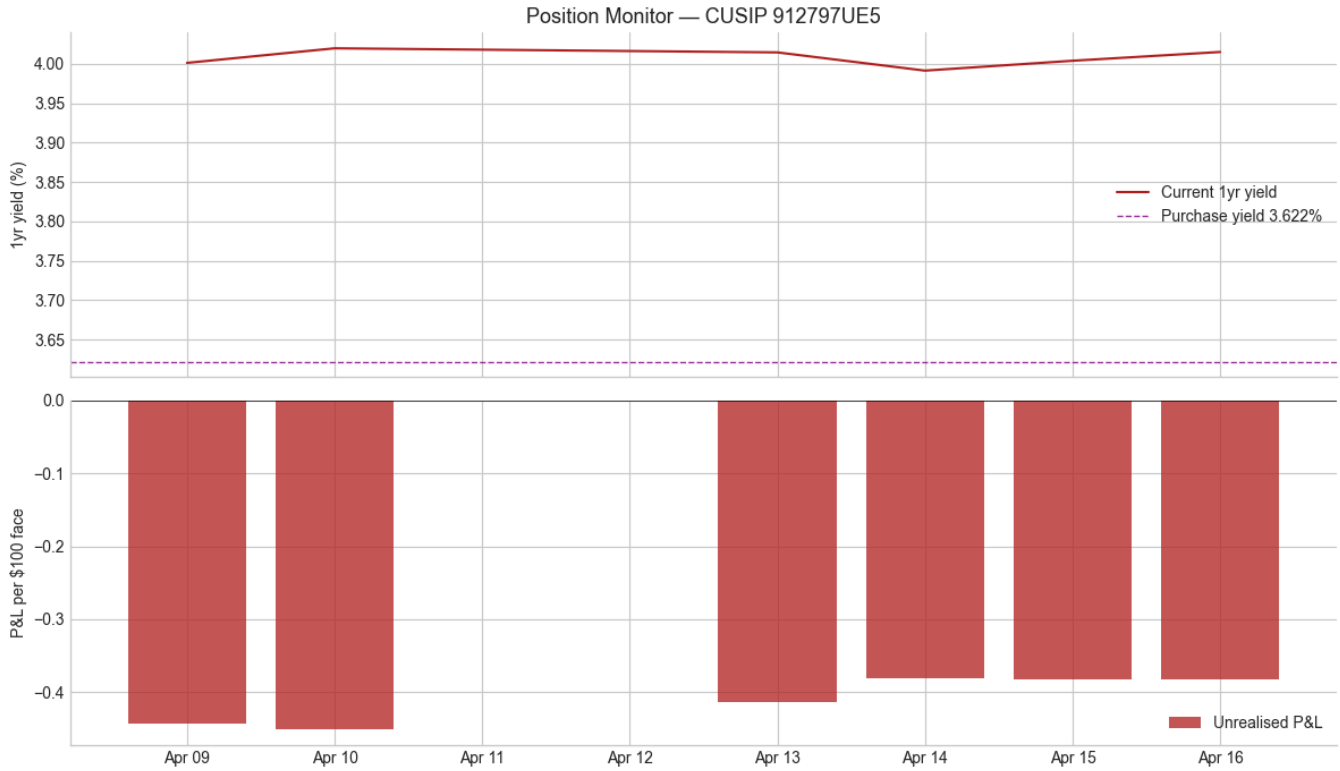
```
Actual in Q10-Q74 band : 76.9%
Avg interval width     : 1.557%
```

4. Current forecast — all horizons



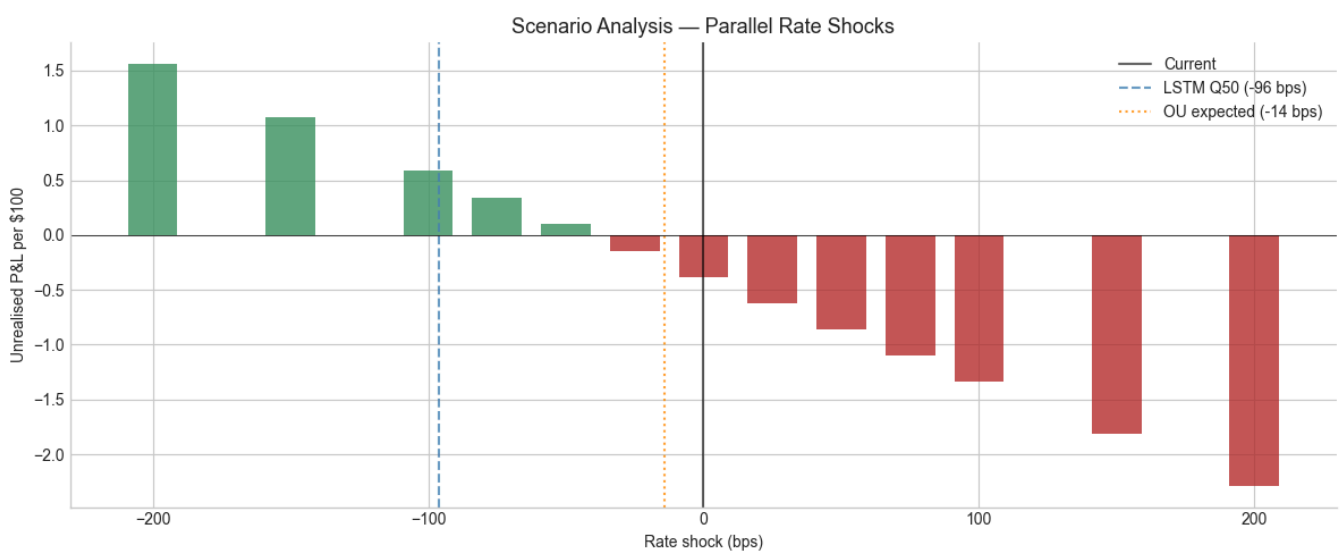
Horizon	Q10	Q50	Q74
21	2.803%	3.111%	3.250%
63	2.253%	2.946%	3.284%
126	2.757%	3.101%	3.399%
252	2.492%	3.054%	3.340%

5. Position P&L since purchase



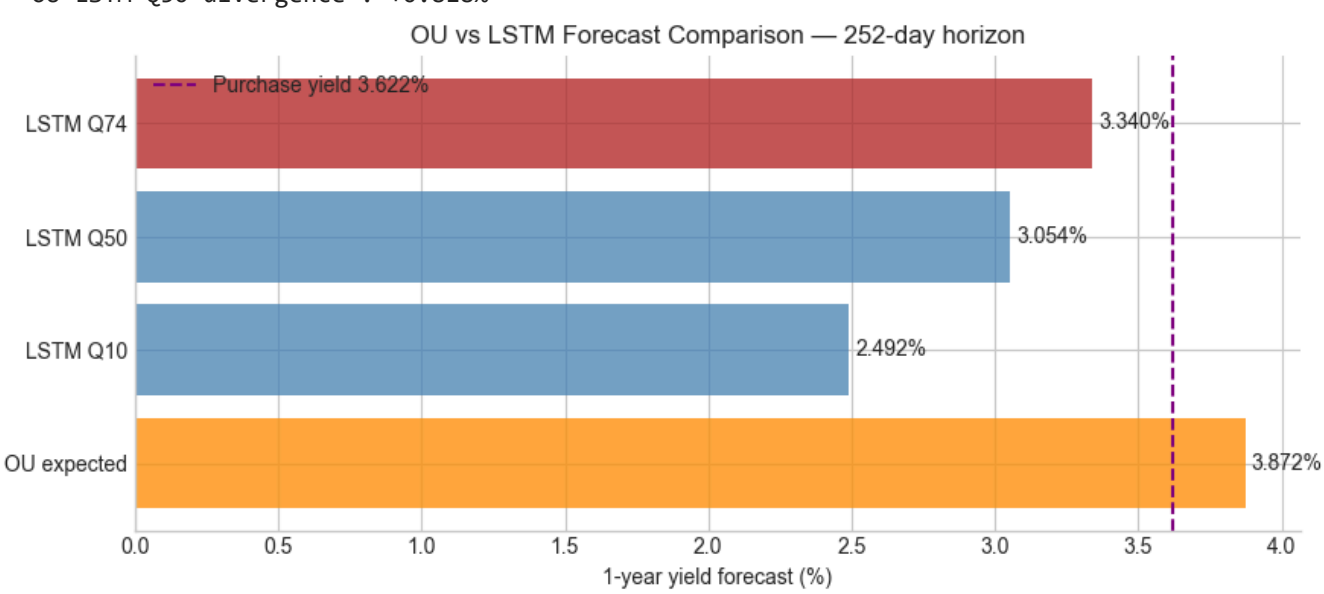

```
Latest date           : 2026-04-16
Current 1yr yield     : 4.015%
Rate change           : +39.0 bps
Unrealised P&L        : $-0.3829
DV01                  : $0.009625
Modified duration     : 1.002 yrs
Current regime        : Tightening
```

6. Scenario analysis



shock_bps	scenario_yield_pct	implied_price	unrealised_pnl	total_pnl
-200	2.0149	98.001184	1.561609	1.561609
-150	2.5149	97.511400	1.071825	1.071825
-100	3.0149	97.024064	0.584489	0.584489
-75	3.2649	96.781310	0.341735	0.341735
-50	3.5149	96.539163	0.099588	0.099588
-25	3.7649	96.297622	-0.141953	-0.141953
0	4.0149	96.056686	-0.382889	-0.382889
25	4.2649	95.816352	-0.623223	-0.623223
50	4.5149	95.576620	-0.862955	-0.862955
75	4.7649	95.337487	-1.102088	-1.102088
100	5.0149	95.098953	-1.340622	-1.340622
150	5.5149	94.623674	-1.815901	-1.815901
200	6.0149	94.150770	-2.288805	-2.288805

7. OU vs LSTM divergence

The two models tell different stories at the 252-day horizon. The OU model forecasts higher rates (tightening regime mean reversion). The LSTM forecasts lower rates (learned regime transition patterns). This divergence reflects genuine forecast uncertainty — both are valid.

```
Forecast Comparison — 252-day horizon
==================================================
  Purchase yield         : 3.622%

  OU expected yield      : 3.872%  +/- 0.183%
  OU P(rates rise)       : 91.4%

  LSTM Q10               : 2.492%
  LSTM Q50 (central)     : 3.054%
  LSTM Q74 (ZCB-optimal) : 3.340%

  OU-LSTM Q50 divergence : +0.818%
```